Dr. Mauricio A. Hess-Flores

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Summary

Computer vision scientist (PhD, Electrical and Computer Engineering) with 20 years of academic, national-laboratory, NASA, and industry experience. Strong background in dense and accurate structure-from-motion from aerial imagery, large-scale photogrammetry, geospatial data processing, and stereo processing for autonomous navigation; also experienced in Computed Tomography (CT) reconstruction and machine learning for object classification. Former lecturer and Lawrence Scholar at LLNL; IEEE Senior Member.

Research Interests

- **Computer vision:** 3D reconstruction, structure-from-motion (SfM), SLAM, epipolar/multi-view geometry, triangulation, photogrammetry, camera calibration, bundle adjustment, pose-graph optimization, geolocation, stereo, mosaicing, tracking, moving object detection, shape-from-X
- **Image processing:** image registration, video stabilization, segmentation, super-resolution, deblurring, denoising, optical flow, feature detection, description, and (learned) matching, change detection
- **Scientific visualization:** visualization of error and uncertainty in computer vision algorithms
- **Inverse problems:** tomography (medical imaging and atmospheric), remote sensing
- **Applied mathematics:** optimization, computational linear algebra, scientific computing, statistical modeling (regression, clustering, model fitting, robust statistics), algebraic geometry/minimal solvers

Education

Ph.D., Electrical and Computer Engineering, University of California, Davis — December 2011
- Dissertation: “Error Detection, Factorization and Correction for Multi-View Scene Reconstruction from Aerial Imagery”
- Advisor: Kenneth I. Joy. Committee: Kenneth I. Joy, Mark A. Duchaineau, Owen T. Carmichael. Minor: Control Systems

M.S., Electrical Engineering (Digital Systems track), with Honors, Universidad de Costa Rica — October 2004
- Thesis: “Automatic adaptation of a human face model to the content of a static image for model-based coding”
- Advisor: Geovanni Martínez

B.S., Electrical Engineering, Universidad de Costa Rica — October 2002

Professional Experience

Principal Scientist – Applied Research — Oct 2025 – present
Spexi Geospatial, Ann Arbor, MI (remote)
Develop computer vision and geospatial algorithms for orthomosaics and 3D products from the world’s largest collection of drone imagery, serving real estate, insurance, and other markets.
- **Production photogrammetry pipeline:** designed, built, and shipped from first principles a drone photogrammetry pipeline (Python, GTSAM, OpenDroneMap, MetaShape, AWS Batch) that turns biased consumer-GPS imagery (2–5 m systematic per-flight error, no RTK/PPK) over thousands of H3 hexagonal cells per year into globally consistent, survey-grade camera poses. Cross-tile alignment is formulated as a global pose-graph optimization solved with GTSAM, achieving sub-meter absolute orthomosaic accuracy on operational data and $<$30 cm on LiDAR-anchored tiles.
- **Basemap anchoring:** built a standalone library using LightGlue deep feature matching against ESRI satellite imagery for sub-meter absolute placement.
- **Quality and triage:** designed a pre-SfM characterization stage that scores each cell on terrain signal, flight geometry, and LiDAR/basemap availability to choose a processing strategy before compute is spent; derived SfM and flight diagnostics enabling automated triage where per-cell human review is impossible.
- **Change detection:** designed a pipeline that suppresses transient noise and quantifies *what* changed and *by how much* across pairs or stacks of orthomosaics (floods, fires, landslides, new construction, vegetation).

Affiliate Scientist — Oct 2025 – present
Space Science Institute, Boulder, CO
- Principal Investigator of a multi-institutional NASA proposal for an open-source framework for mission-spanning 3D reconstruction and geometric calibration of Europa using Voyager, Galileo, and Juno imagery.
- Developing research directions in planetary computer vision, multi-mission image registration, 3D reconstruction, and scientific data products.

Data Scientist / Geospatial Image Data Specialist (Caltech Key Staff) Jan 2021 – Oct 2025
NASA Jet Propulsion Laboratory / Caltech, Pasadena, CA

- **Mars 2020 and multi-mission ground processing:** developed a SLAM algorithm for in-situ imagery in the VICAR suite, improving performance on Ingenuity flight paths with tight corners (2021–2022); delivered further VICAR vision tools such as 3D reconstruction from depth-from-focus (2022); Cognizant Engineer for the Mars 2020 raw data processing software, edrgen (2022–2024).
- **Earth science geolocation:** delivered geolocation tools and data for the CALET instrument on the ISS in support of OCO-3, improving geolocation from kilometers to hundreds of meters, plus Python tools for cross-instrument geolocation (e.g., ECOSTRESS) (2021–2023). Established a strategic collaboration with the Surface Biology and Geology team on AVIRIS-NG geolocation for the SHIFT campaign, contributing to datasets published in the ORNL DAAC (2022–2025).
- **Error visualization:** PI of a Caltech Data-to-Discovery award; led a team of interns to build the VECTOR error-visualization tool, now used in Mars 2020 operations (IEEE VIS 2024) (2022–2023).
- **NASA GIBS / Worldview:** implemented data compression and browse-image generation for NASA's Global Imagery Browse Services, whose imagery is cited in over 300 publications per year (2023–2025).
- **Atmospheric tomography:** developed algorithms for cloud and water-vapor tomography for the NASA Earth Science Technology Office (2024).
- **Other:** delivered video stabilization and image registration software, written from scratch, to internal customers (2022–2023); member of the Habitable Worlds Observatory AI/ML Working Group (2025); public outreach at the high-school and college level.

Senior Engineer / Research Scientist Jan 2017 – Dec 2020
Stratovan Corporation, Davis, CA

- Developed novel CT reconstruction algorithms with the founders (Dr. David Wiley and Prof. Bernd Hamann) that reduce streak and metal artifacts in medical and airport-security applications, including large matrix inversion methods.
- Completed a feature-selection and optimization pipeline for SVM-based object classification in airport security.
- Investigated shape analysis and feature extraction for dental radiology; organized a video series with academic collaborators showcasing the company's Checkpoint capabilities.

Lead Scientist Jul 2016 – Oct 2016
uSens, Inc., San Jose, CA

- Investigated and tested SLAM approaches for augmented and virtual reality (AR/VR) applications.

Senior Engineer, Corporate R&D Computer Vision Nov 2014 – Jul 2016
Qualcomm Research Center, San Diego, CA

- Developed 3D computer vision software for embedded devices.
- Significantly improved the accuracy of the group's stereo algorithm, the world's best performing per the Middlebury benchmark (2015).
- Designed and implemented stereo-based ground-plane segmentation and collision detection for autonomous vehicles.

Postdoctoral Scholar Jan 2012 – Oct 2014
Institute for Data Analysis and Visualization (IDAV), University of California, Davis

- Led computer vision research at IDAV and mentored PhD students; authored 15+ publications on multi-view scene reconstruction from aerial imagery.
- Developed novel cost functions for accurate and efficient multi-view triangulation, and visualization techniques for computer vision analysis.

Lawrence Scholar Program Fellow Apr 2009 – Nov 2011
Lawrence Livermore National Laboratory, Livermore, CA

- Joint IDAV–LLNL collaboration designing, implementing, and publishing algorithms for multi-view reconstruction from aerial imagery; algorithms became part of a laboratory-level reconstruction pipeline.
- Delivered dozens of technical talks on literature reviews and new algorithm design.

Graduate Student Researcher Sep 2007 – Mar 2009
IDAV, University of California, Davis (advisor: Kenneth I. Joy)

- Worked with LLNL researchers on publications and systems for multi-view reconstruction from aerial imagery.

Summer Scholar Summers 2007 and 2008
Lawrence Livermore National Laboratory, Livermore, CA

- Multi-view reconstruction from aerial imagery under Mark A. Duchaineau, Institute for Scientific Computing Research.

Product Engineer Jul 2004 – Jul 2006
TriQuint Semiconductor, Heredia, Costa Rica
- Coordinated statistical process control for the Costa Rica site, reporting to headquarters in Orlando, FL; yield-improvement lead for 75+ part numbers in production.

Research Assistant 2004
Image Processing and Computer Vision Laboratory, Universidad de Costa Rica
- Research on image processing and model-based coding with Dr. Geovanni Martínez, leading to two publications.

Research Proposals

Principal Investigator, NASA ROSES-2025 (C.2 SCUBED), Space Science Institute (2026)
"Establishing a Stereo-Processor for Europa Through Calibrated Cross-Mission Images." A mission-spanning stereo reconstruction and geometric calibration framework for Voyager, Galileo, and Juno Europa imagery, with open-source software, 3D surface products, uncertainty maps, and PDS-compatible data products. Reviewed with zero Major Technical Weaknesses; rated "high" for technical feasibility and pipeline scalability.

Co-Principal Investigator, NNSA (PI: Kenneth I. Joy)
"Parallax Paths Framework for Accurate, Dense Scene Reconstruction from Aerial Video."

Honors and Awards

- NASA ESDIS Worldview/GIBS Robert H. Goddard Exceptional Achievement for Engineering Team Award (2024)
- NASA Global Imagery Browse Services (GIBS) Team Bonus Award (2024)
- OCO-3 Team Bonus Awards (2022, 2023)
- JPL Voyager Award (2022), for the Data-to-Discovery proposal and the AVIRIS-NG/SHIFT geolocation collaboration
- Caltech Data to Discovery proposal award (2022)
- Best Paper Award, CONTENT 2014 (Venice, Italy)
- International Computer Vision Summer School (2013): one of 140 graduate students and postdocs selected worldwide
- Lawrence Scholar Program Fellowship, Lawrence Livermore National Laboratory (2009)
- Office of Graduate Studies Fellowship, University of California, Davis (2006)
- M.S. graduation with Honors (GPA $\geq$ 9.0/10) (2004) and Honors Scholarship (1998), Universidad de Costa Rica

Publications

Journal Articles

- L. Forster, M. T. Richardson, A. B. Davis, **M. A. Hess-Flores**, I. Yanovsky, J. Ryan, M. D. Lebsock, D. J. Posselt, F. C. Seidel, C. O. Ao, S. T. Brown, D. J. Diner, E. F. Fishbein, L. Di Girolamo, M. J. Kurowski, L. Mandrake, C. E. Miller, N. C. Parazoo, G. L. Stephens, J. P. Teixeira, J. J. Hyon, M. S. Seablom, "Scientific Applications of Multi-Angle Measurements to Reconstruct Earth's Atmosphere through Tomography," *Bulletin of the American Meteorological Society*, 2026. doi:10.1175/BAMS-D-24-0293.1

Peer-Reviewed Conference and Workshop Papers

- R. Fygenson, K. Jawad, I. Li, F. Ayoub, R. G. Deen, S. Davidoff, D. Moritz, **M. A. Hess-Flores**, "Opening the black box of 3D reconstruction error analysis with VECTOR," *IEEE Visualization and Visual Analytics (VIS)*, pp. 286–290, 2024.
- J. Mak, **M. A. Hess-Flores**, S. Recker, J. D. Owens, K. I. Joy, "A Comparative Study of Recent GPU-Accelerated Multi-View Sequential Reconstruction Triangulation Methods for Large-Scale Scenes," *ACCV Workshop: Big Data in 3D Computer Vision*, Singapore, 2014.
- S. Recker, C. Gribble, M. M. Shashkov, M. Yepez, **M. A. Hess-Flores**, K. I. Joy, "Depth Data Assisted Structure-from-Motion Parameter Optimization and Feature Track Correction," *IEEE Applied Imagery Pattern Recognition Workshop (AIPR)*, Washington, DC, 2014.
- **M. A. Hess-Flores**, S. Recker, K. I. Joy, "Uncertainty, Baseline, and Noise Analysis for L_1 Error-Based Multi-View Triangulation," *International Conference on Pattern Recognition (ICPR)*, Stockholm, Sweden, 2014.
- M. M. Shashkov, C. Nguyen, M. Yepez, **M. A. Hess-Flores**, K. I. Joy, "Semi-Autonomous Digitization of Real-World Environments," *International Conference on Computer Games (C-GAMES USA)*, Louisville, KY, 2014.
- S. Recker, M. M. Shashkov, **M. A. Hess-Flores**, C. Gribble, R. Baltrusch, M. A. Butkiewicz, K. I. Joy, "Hybrid Photogrammetry Structure-from-Motion Systems for Scene Measurement and Analysis," *Coordinate Metrology Systems Conference (CMSC)*, North Charleston, SC, 2014. *Featured in Quality Digest, November 2014.*
- M. M. Shashkov, **M. A. Hess-Flores**, S. Recker, K. I. Joy, "Towards Sensor-Aided Multi-View Reconstruction for High Accuracy Applications," *International Conference on Creative Content Technologies (CONTENT)*, Venice, Italy, 2014. **Best Paper Award.**
- J. Mak, **M. A. Hess-Flores**, S. Recker, J. D. Owens, K. I. Joy, "GPU-Accelerated and Efficient Multi-View Triangulation for Scene

Reconstruction," *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2014.
- S. Recker, **M. A. Hess-Flores**, K. I. Joy, "Fury of the Swarm: Efficient and Very Accurate Triangulation for Multi-View Scene Reconstruction," *ICCV Workshop: Big Data in 3D Computer Vision*, Sydney, Australia, 2013.
- S. Recker, **M. A. Hess-Flores**, K. I. Joy, "Feature Track Summary Visualization for Multi-View Reconstruction," *IEEE AIPR Workshop*, 2013.
- **M. A. Hess-Flores**, S. Recker, K. I. Joy, M. A. Duchaineau, "Visualization Methods for Computer Vision Analysis," *International Conference on Pervasive Patterns and Applications (PATTERNS)*, Valencia, Spain, 2013.
- S. Recker, **M. A. Hess-Flores**, K. I. Joy, "Statistical Angular Error-Based Triangulation for Efficient and Accurate Multi-View Scene Reconstruction," *IEEE WACV*, pp. 68–75, 2013.
- **M. A. Hess-Flores**, M. A. Duchaineau, K. I. Joy, "Sequential Reconstruction Segment-Wise Feature Track and Structure Updating Based on Parallax Paths," *Asian Conference on Computer Vision (ACCV)*, Springer, pp. 636–649, 2012.
- S. Recker, **M. A. Hess-Flores**, M. A. Duchaineau, K. I. Joy, "Visualization of Scene Structure Uncertainty in a Multi-View Reconstruction Pipeline," *Vision, Modeling and Visualization Workshop (VMV)*, pp. 183–190, 2012.
- **M. A. Hess-Flores**, M. A. Duchaineau, K. I. Joy, "Path-Based Constraints for Accurate Scene Reconstruction from Aerial Video," *IEEE AIPR Workshop*, 2012.
- S. Recker, **M. A. Hess-Flores**, M. A. Duchaineau, K. I. Joy, "Visualization of Scene Structure Uncertainty in Multi-View Reconstruction," *IEEE AIPR Workshop*, 2012.
- **M. A. Hess-Flores**, D. Knoblauch, M. A. Duchaineau, K. I. Joy, F. Kuester, "Ray Divergence-Based Bundle Adjustment Conditioning for Multi-View Stereo," *Pacific-Rim Symposium on Image and Video Technology (PSIVT)*, Springer LNCS 7087, pp. 153–164, 2011.
- D. Knoblauch, **M. A. Hess-Flores**, M. A. Duchaineau, K. I. Joy, F. Kuester, "Non-Parametric Sequential Frame Decimation for Scene Reconstruction in Low-Memory Streaming Environments," *International Symposium on Visual Computing (ISVC)*, Springer LNCS 6938, pp. 363–374, 2011.
- **M. A. Hess-Flores**, M. A. Duchaineau, M. J. Goldman, K. I. Joy, "Iterative Dense Correspondence Error Correction Through Bundle Adjustment Feedback-Based Error Detection," *International Conference on Computer Vision Theory and Applications (VISAPP)*, pp. 400–405, 2010.
- D. Knoblauch, **M. A. Hess-Flores**, M. A. Duchaineau, F. Kuester, "Factorization of Correspondence and Camera Error for Unconstrained Dense Correspondence Applications," *ISVC*, pp. 720–729, 2009.
- **M. Hess**, G. Martínez, "Automatic Adaptation of a Human Face Model for Model-Based Coding," *Picture Coding Symposium (PCS)*, San Francisco, CA, 2004.
- **M. Hess**, G. Martínez, "Facial Feature Extraction Based on the Smallest Univalue Segment Assimilating Nucleus (SUSAN) Algorithm," *Picture Coding Symposium (PCS)*, San Francisco, CA, 2004.

Conference Presentations and Abstracts

- **M. A. Hess-Flores**, J. Forsyth, G. Hemann, "Scalable High-Accuracy Drone Mapping for Urban Digital Twins: A Tiled Reconstruction Pipeline," *AGU Fall Meeting*, San Francisco, CA, December 2026, GC31F-03 (oral, accepted).
- **M. A. Hess-Flores**, J. Forsyth, G. Hemann, "GeoAI for City-Scale Drone Mapping Using Hierarchical Spatial Indexing and Global Pose Optimization," *AGU Fall Meeting*, San Francisco, CA, December 2026 (poster, accepted).
- L. Forster, M. Richardson, **M. A. Hess-Flores**, I. Yanovsky, A. B. Davis, et al., "New Angles on the PBL: Potential and Limitations of Space-borne Atmospheric Tomography for Characterizing the Planetary Boundary Layer," *AGU Fall Meeting*, Washington, DC, 2024, A24G-07 (oral); also presented at the 105th AMS Annual Meeting, New Orleans, LA, 2025.
- **M. A. Hess-Flores**, J. T. Roberts, M. W. Graber, J. Ryan, O. Pariser, G. Llewellyn, Z. Xing, L. Plesea, M. Wong, R. A. Boller, A. Li, "Storage Optimization for NASA's Massive Archive of Earth Imagery and Visualized Data – NASA's Global Imagery Browse Services," *AGU Fall Meeting*, Washington, DC, 2024 (poster). ESS Open Archive, doi:10.22541/essoar.173834702.29875040/v2

Data Products

P. G. Brodrick, R. Pavlick, M. Bernas, J. W. Chapman, R. Eckert, M. Helmlinger, **M. A. Hess-Flores**, et al., SHIFT airborne campaign datasets, ORNL DAAC, Oak Ridge, TN:

- AVIRIS-NG L1B Calibrated Radiance, V2 (2025). doi:10.3334/ORNLDAAC/2432
- AVIRIS-NG L2A Orthorectified Surface Reflectance, V2 (2025). doi:10.3334/ORNLDAAC/2431
- AVNG L2 Weekly Georeferenced Surface Reflectance Mosaics (2025). doi:10.3334/ORNLDAAC/2289
- AVIRIS-NG L2A Unrectified Surface Reflectance, V1 (2023). doi:10.3334/ORNLDAAC/2376
- AVIRIS-NG L1A Unrectified Radiance (2023). doi:10.3334/ORNLDAAC/2184
- AVIRIS-NG L2A Unrectified Reflectance (2023). doi:10.3334/ORNLDAAC/2183
- AVIRIS-NG Full-Resolution True Color Images (2023)

Invited Talks and Posters

- **M. A. Hess-Flores**, K. I. Joy, "Real-Time 3D Exploration of Wide Area High-Resolution Aerial Video," DOE University and Industry Technical Interchange (UITI), Lansing, MI, June 2013 (talk).

- J. Mak, **M. A. Hess-Flores**, S. Recker, J. D. Owens, K. I. Joy, "Multi-View 3D Aerial Reconstruction Algorithms on the GPU," NVIDIA GPU Technology Conference, San Jose, CA, March 2013 (poster).
- **M. A. Hess-Flores**, "Large-scale reconstruction from aerial images," LLNL Lawrence Scholar Program Poster Session (2009) and Summer Student Poster Sessions (2007, 2008), Livermore, CA.

Teaching Experience

Assistant Professor (Lecturer) Mar 2005 – Jul 2006
Graduate Program in Electrical Engineering, Universidad de Costa Rica
- Lecturer for graduate courses in Probability, Random Variables and Stochastic Processes, and the Digital Systems Laboratory (microcontrollers and DSPs).
- Thesis committee member for one Bachelor's and one Master's thesis.

Teaching Assistant Sep 2006 – Jun 2007
Department of Electrical and Computer Engineering, University of California, Davis
- Circuits I (ENG17), Fall 2006 and Spring 2007; Engineering Problem-Solving (ENG6), Winter 2007.

Teaching Assistant 2001 – 2002
School of Physics, Universidad de Costa Rica

Students Mentored

- **University of California, Davis (PhD, Computer Science):** Shawn Recker and Mikhail M. Shashkov (advisor: Kenneth I. Joy); Jason Mak (advisor: John D. Owens)
- **NASA JPL** / **Caltech:** intern team that developed the VECTOR error-visualization tool (2022–2023)
- **Universidad de Costa Rica (committee member):** Javier Rivera Alvarado, M.S. thesis, "Desarrollo de un sistema experto para la automatización del análisis de fallas de filtros RF de tecnología SAW"; Sergio Oses Pushkov, B.S. project, "Estudio sobre Sistemas de Posicionamiento"

Professional Service

- **Mentorship:** Invited Mentor, IEEE CVPR Speed Mentorship Program, Denver, CO (2026), one of 18 industry/academic leaders selected by IEEE to mentor top computer vision students (180 selected from 1,500+ applicants)
- **Conference reviewer:** CVPR (2020–2025); ECCV (2022–2024); ICCV (2021); WACV (2015–2022, 2024); BMVC (2026); IEEE Aerospace Conference (2022); IWOBI (2018); CARLA (2019); CGVCVIP (2012, 2014, 2016, 2021); PATTERNS (2013–2015)
- **Program committees:** Advisory Chair, PATTERNS (2014, 2015); Technical Program Committee, PATTERNS (2013–2015); IADIS CGVCVIP (2012–2014, 2016, 2021); CARLA 2019, Bioinspired Processing track
- **University service:** Judge, Interdisciplinary Graduate and Professional Student Symposium, UC Davis (2012–2014)

Professional Affiliations

- Institute of Electrical and Electronics Engineers (IEEE): Senior Member (since 2024); Geoscience and Remote Sensing Society (since 2024); Robotics and Automation Society (2014–2023)
- American Geophysical Union (since 2023); Computer Vision Foundation (since 2013); Costa Rican Association of Engineers and Architects (since 2002)

Technical Skills

- **Languages:** Python, C/C++, Matlab/Octave, Unix shell
- **Vision, geospatial, and optimization:** GTSAM, OpenCV, Eigen, LightGlue, OpenDroneMap, Agisoft MetaShape, H3, VICAR, SBA/Levmar
- **Infrastructure and tools:** AWS Batch, Git, Perforce; Linux/Unix, macOS, Windows